

Data-Driven Insights for Kalka Dairy Farm: Analysis Process, Findings, and Strategic Recommendations

A Final Report For the BDM Capstone Project

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1. Executive Summary –

Kalka Dairy Farm is a small dairy business that manages milk production, animal health, customer sales, feed consumption and expenses. The project was undertaken to address the limitations of manual and scattered record-keeping and to develop a structured Business Data Management approach for improving operational and financial decision-making. The study used data from January to June 2025 and applied data cleaning, descriptive analysis, trend analysis, cost analysis, profitability analysis, KPI analysis and dashboard-based visualization using Excel and Python. These methods were selected to convert routine farm records into meaningful business insights.

The final results show that the farm maintained a stable and profitable performance. Total reported milk production was 6,930 litres, revenue was ₹4,17,800, total expenses were ₹2,00,320, and net profit was ₹2,17,480, giving an overall profit margin of approximately 52.1%. Production improved after the February low and reached its strongest level in May, while June remained comparatively strong.

The project recommends maintaining daily digital records, monitoring animal-wise production and feed efficiency, controlling operating expenses, following up pending customer payments, maintaining preventive animal-health records, and reviewing KPIs monthly. Overall, the project demonstrates that structured data management and simple analytics can help Kalka Dairy Farm protect profitability, identify operational issues earlier, and make evidence-based decisions for sustainable growth.

2. Detailed Explanation of Analysis Process/Method –

2.1 Overview of the Analysis Method

The analysis of Kalka Dairy Farm was conducted using a structured, data-driven approach to understand milk production, sales, customer transactions, feed consumption, animal health, operating expenses and profitability. The method was selected to directly address the farm's business problem of maintaining scattered/manual records and difficulty in monitoring performance. **dataset collected from January 2025 to June 2025(6 Months).**

The complete process followed the sequence: Data Collection → Data Preparation → Data Validation → Descriptive Statistics → Production Analysis → Sales Analysis → Cost Analysis → Profitability Analysis → KPI Analysis → Visualization → Interpretation → Recommendations.

2.2 Data Collection and Dataset Structure

The farm data was organized into separate but related Excel datasets. The main datasets include Animal_Master, Daily_Milk_Production, Customers, Milk_Sales, Feed_Consumption, Veterinary_Records, Expenses, Monthly_Summary, Dashboard_Data and KPI/KPI Metadata. Together, these datasets represent the major operational and financial activities of the dairy farm.

Animal_Master – animal identification and basic animal information.

Daily_Milk_Production – daily morning, evening and total milk production.

Customers and Milk_Sales – customer details and milk-selling transactions.

Feed_Consumption – feed quantity, type and cost.

Veterinary_Records – treatment, health and veterinary cost information.

Expenses – operational expenditure records.

Monthly_Summary and KPI – summarized performance indicators for reporting.

2.3 Data Preparation, Cleaning and Validation

Before analysis, the datasets were checked for missing values, duplicate records, incorrect data types, inconsistent dates, categorical inconsistencies and numerical-format errors. Date fields were standardized, while quantities and financial values were treated as numerical variables. Categorical variables such as animal type, customer type and payment mode were standardized for grouping.

A reconciliation step was also included to compare detailed transaction-level records with summary tables. This prevents incorrect summary figures from being used without checking their relationship with the underlying records. Data validation is particularly important because manual record keeping was identified as a business challenge for the farm.

2.4 Descriptive Statistical Analysis

Descriptive statistics were used to understand the distribution and variation of important numerical variables. The analysis includes count, total, mean, median, minimum, maximum, standard deviation and quartiles (25%, 50% and 75%). These measures were applied to milk production, milk sales, selling price, feed consumption, feed cost, veterinary cost, expenses, revenue, net profit and profit margin.

This method is appropriate because the first requirement of the project is to understand the current state of the farm. Mean and median show typical performance, while minimum, maximum and standard deviation reveal the amount of variation in the records.

2.5 Milk Production Analysis

Milk production was analysed at both animal and monthly levels. Morning milk, evening milk and total daily milk were examined using the relationship: $\text{Total Milk} = \text{Morning Milk} + \text{Evening Milk}$. Production was grouped by animal and month to identify high- and low-performing animals, monthly changes, peaks and declines. Line charts and bar charts were used to make these patterns visible.

This method is appropriate because milk production is the primary output of the dairy farm. Animal-wise analysis helps identify production differences, while monthly analysis shows whether overall output is improving or declining.

2.6 Sales and Customer Analysis

The Milk_Sales dataset was analysed to understand the farm's revenue-generating activity. The analysis considered quantity sold, rate per litre, transaction amount, customer, payment mode and payment status. The dataset contains 1,086 sales transactions, with an average quantity of approximately 4.51 litres per transaction and an average transaction value of approximately ₹265.79.

The selling rate was also examined. Since the rate is concentrated around ₹60 per litre, changes in revenue can be studied mainly through quantity sold and transaction volume. The Customers dataset, with 40 customer records, supports customer-wise and customer-type comparisons.

This method is appropriate because total revenue alone cannot explain sales performance. Combining quantity, price, customer and transaction information provides a clearer explanation of income changes.

2.7 Feed Consumption and Cost Analysis

Feed consumption was analysed using quantity, feed type, cost per kilogram, total feed cost, animal and supplier. The Feed_Consumption dataset contains 1,448 records. Average feed consumption is approximately 13.08 kg per record and average total feed cost is approximately ₹141.75.

Feed usage was considered together with milk production to understand whether higher input consumption is associated with higher output. This is appropriate because feed is a recurring production input and its cost can have a direct effect on profitability.

2.8 Veterinary and Animal Health Analysis

Veterinary_Records contains 74 treatment records. Treatment type, medicine, treatment cost, health status, veterinarian and next visit were examined. Average treatment cost is approximately ₹629.05 per record. Where appropriate, health records can be compared with animal production to identify possible relationships between health events and productivity.

This method is suitable because animal health can influence both milk output and operating costs. Including veterinary data therefore gives a more complete view of farm performance.

2.9 Operating Expense Analysis

The Expenses dataset contains 724 records. Expenses were examined by category, amount and time period. The analysis calculates total, average, minimum, maximum and standard deviation of expenses and compares major categories such as feed, labour, veterinary, electricity, maintenance and other operating costs.

The average expense record is approximately ₹982.75, with considerable variation between entries. Category-wise charts help identify the major sources of expenditure and areas requiring closer monitoring.

2.10 Profitability Analysis

Financial performance was evaluated by comparing revenue and expenses. Net Profit was calculated as Revenue – Total Expenses, while Profit Margin was calculated as $(\text{Net Profit} \div \text{Revenue}) \times 100$. Monthly revenue, expenses, profit and margin were compared to identify stronger and weaker periods.

This method is appropriate because higher sales do not necessarily result in higher profit. Comparing income with costs provides a more meaningful measure of financial performance.

2.11 Trend Analysis

Trend analysis was applied to milk production, sales quantity, revenue, expenses, net profit and profit margin over the six-month period. Line and column charts were used to identify increases, decreases, peaks and unusual movements. The project results indicate a weaker period around February followed by improvement, with May showing particularly strong production and financial performance.

2.12 KPI and Dashboard Analysis

Key Performance Indicators were used to provide a concise view of the farm's performance. Important KPIs include total milk produced, total milk sold, total revenue, total expenses, net profit, profit margin, number of animals and number of customers. These indicators are presented through KPI cards, summary tables and charts.

The dashboard approach is appropriate because the farm owner should be able to monitor important indicators without examining every individual transaction. It converts detailed records into a quick decision-support tool.

2.13 Visualization Method

Graphs were used to communicate the results and make trends easier to interpret. Recommended visuals include monthly milk-production line charts, animal-wise production bar charts, revenue-versus-expense charts, monthly profit charts, profit-margin charts, customer-wise sales charts, feed-cost charts, expense-category charts and a KPI dashboard.

Visualization is not used only for presentation. It helps identify patterns, peaks, declines and differences that may be difficult to recognize from tables alone.

2.14 SWOT and Root Cause Analysis

Supporting business-analysis methods were used to convert statistical findings into practical business insights. SWOT Analysis considers strengths, weaknesses, opportunities and threats, while Root Cause Analysis examines why an identified problem occurs.

Strengths: established dairy operations, regular production and an existing customer base.

Weaknesses: manual/scattered records and limited centralized reporting.

Opportunities: digital records, KPI monitoring and improved production-cost analysis.

Threats: feed-price changes, animal-health issues and production fluctuations.

For example, difficulty in monitoring profitability can be traced to separate records, manual calculations and delayed reporting. A centralized dataset with automated calculations and KPI reporting can address these causes.

2.15 Tools Used for Analysis

Microsoft Excel was used for data organization, formulas, Pivot Tables, charts, KPI summaries and dashboard preparation. Python/Google Colab was used for data loading, cleaning, grouping, statistical analysis and visualization. Using both tools provides business-friendly reporting through Excel and reproducible analytical processing through Python.

2.16 Why This Method Is More Appropriate Than Other Methods

The selected methodology is appropriate because the immediate objective is to understand the farm's existing performance, identify operational problems and provide practical recommendations. Descriptive statistics explain the current data; trend analysis explains changes over time; production, sales and cost analysis identify operational drivers; profitability analysis evaluates financial performance; and KPI dashboards support regular monitoring.

Advanced techniques such as machine-learning prediction, ARIMA forecasting or optimization models were not selected as the primary methods because the project first requires reliable understanding of the existing data. Such techniques could be considered in a future study after the data-management system and historical records become more extensive.

2.17 Overall Analysis Framework

**Raw Farm Data → Cleaning & Validation → Descriptive Statistics → Production Analysis
→ Sales & Customer Analysis → Feed & Veterinary Analysis → Expense Analysis
→ Profitability & KPI Analysis → Visualization → Interpretation → Recommendations**

Overall, this methodology transforms Kalka Dairy Farm's operational records into structured information for monitoring production, controlling costs, understanding sales performance, evaluating profitability and supporting better business decisions. The approach is simple, transparent, reproducible and directly connected to the farm's identified business problems.

3. Results and Findings –

This section presents the major findings from the Kalka Dairy Farm dataset for January–June 2025. The results focus on milk production, revenue, expenditure and profitability. Graphical representations are used to make the operational and financial trends easy to understand.

3.1 Overall Performance Snapshot

Key Indicator	Result
Total milk produced	6,930 L
Average monthly production	1,155 L
Average daily production	38.5 L/day
Total revenue	₹415,800
Total expenses	₹200,320
Total net profit	₹215,480
Average profit margin	51.81%
Feed cost share	56.57% of expenses

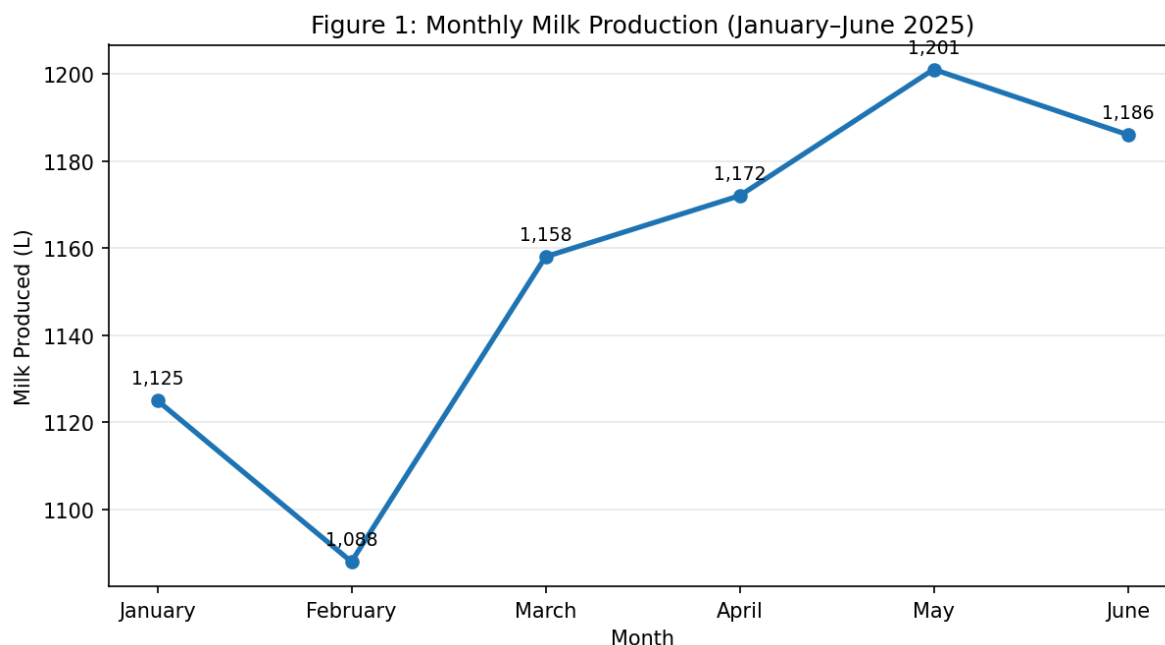
The farm produced 6,930 litres during the six-month period, averaging 1,155 litres per month. Average daily production was 38.5 litres. Total revenue exceeded total expenses, resulting in a positive consolidated net profit of ₹215,480.

3.2 Key Findings

- Production increased from 1,125 L in January to 1,186 L in June, a 5.4% net rise.
- February was the lowest production month at 1,088 L; May was highest at 1,201 L.
- May also recorded the highest revenue (₹72,060) and net profit (₹37,480).
- Profit was positive in all six months, with an average margin of 51.81%.
- The overall pattern is stable with a noticeable improvement after February.

These results directly address the project objective of converting scattered business records into a structured view of farm performance. The proposal identifies manual record management and limited business monitoring as important challenges; the analysis demonstrates how the same records can be compared systematically.

3.3 Monthly Milk Production Trend



Month	Milk (L)	Change
Jan	1,125	—
Feb	1,088	-3.3%
Mar	1,158	+6.4%
Apr	1,172	+1.2%
May	1,201	+2.5%
Jun	1,186	-1.2%

Interpretation

Milk production fell from 1,125 L in January to 1,088 L in February. It then recovered through March and April and reached a peak of 1,201 L in May. June remained strong at 1,186 L, although it was slightly below May. From February to May, production increased by 10.4%.

The February dip is therefore the main low point in the series. The subsequent rise indicates an improvement in output during the middle and later months of the study period. Because milk is the farm's primary product, production is an important leading indicator for sales and revenue.

Animal Productivity

The Animal_Master worksheet contains 8 records. Average recorded animal productivity is 10.78 L/day. The highest average is recorded for Radha (12.3 L/day), while the lowest is Ganga (8.8 L/day).

The variation between animals suggests that animal-wise production monitoring can help management identify higher- and lower-performing animals and then compare their feed and health records. This is more informative than relying only on a farm-wide average.

3.4 Revenue, Expenses and Profitability

Financial results across six months

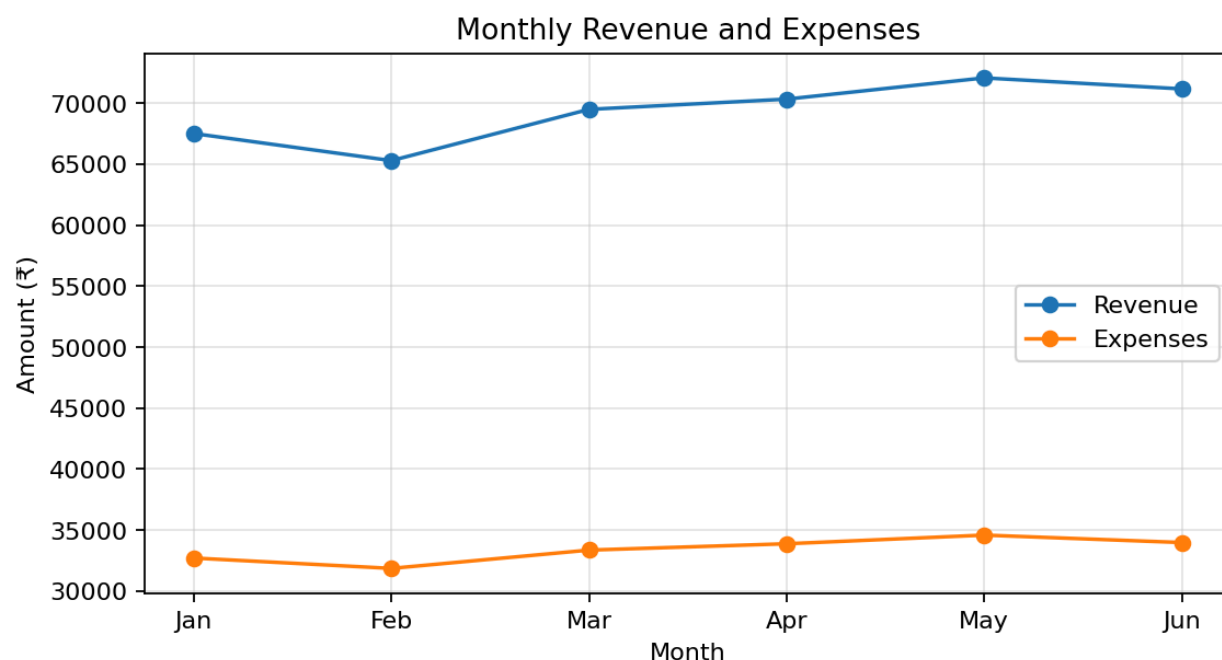


Figure 2. Monthly revenue compared with reported expenses.

Month	Revenue (₹)	Expenses (₹)	Profit (₹)	Margin
Jan	67,500	32,700	34,800	51.56%
Feb	65,280	31,850	33,430	51.21%
Mar	69,480	33,350	36,130	52.00%
Apr	70,320	33,870	36,450	51.83%
May	72,060	34,580	37,480	52.01%
Jun	71,160	33,970	37,190	52.26%

Findings

- Revenue rose from ₹67,500 in January to a peak of ₹72,060 in May.
- From February to May, revenue increased by 10.4%.
- Total revenue of ₹415,800 exceeded total reported expenses of ₹200,320.

- Total net profit across the six months was ₹215,480.
- Average monthly net profit was ₹35,913.33.

Revenue and profit generally moved in the same direction as production. This suggests that stronger output and sales activity were associated with better financial returns during the observed period. However, revenue alone is not sufficient to judge performance; expenses must be considered at the same time.

The relatively consistent gap between revenue and expenses produced positive monthly profit. The strongest financial month was May, while February was the weakest. This makes monthly comparison a useful management practice for identifying deviations early.

3.5 Profit and Profit Margin Findings

Stability of financial performance

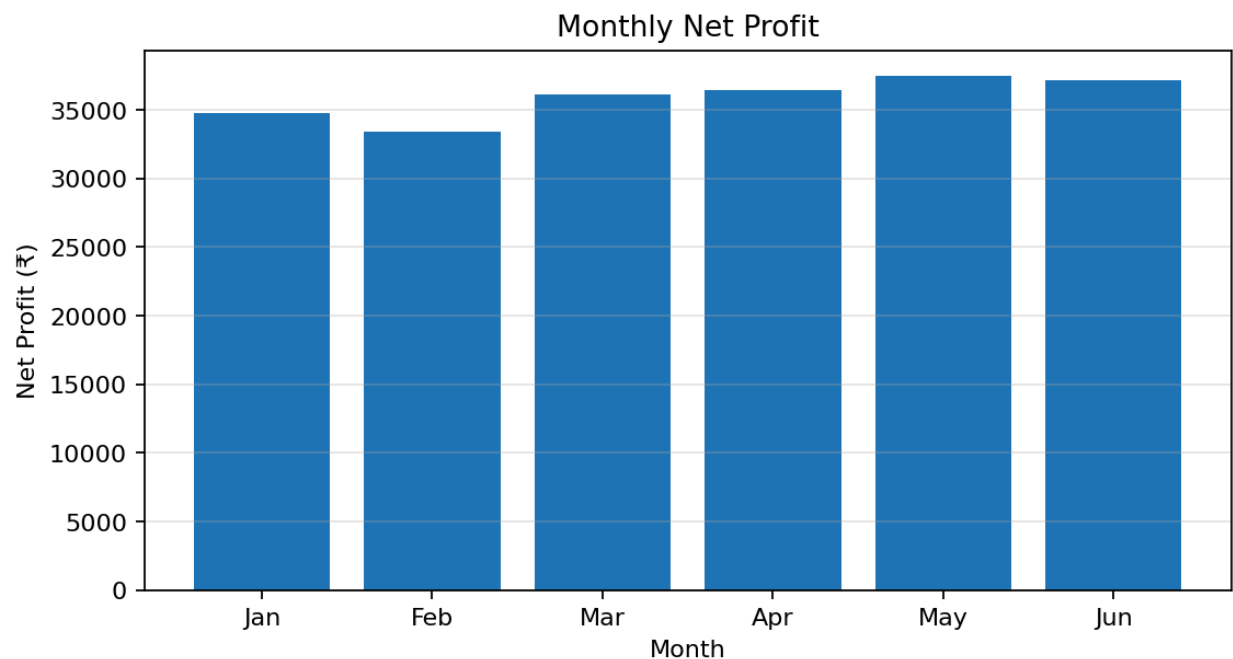


Figure 3. Monthly net profit.

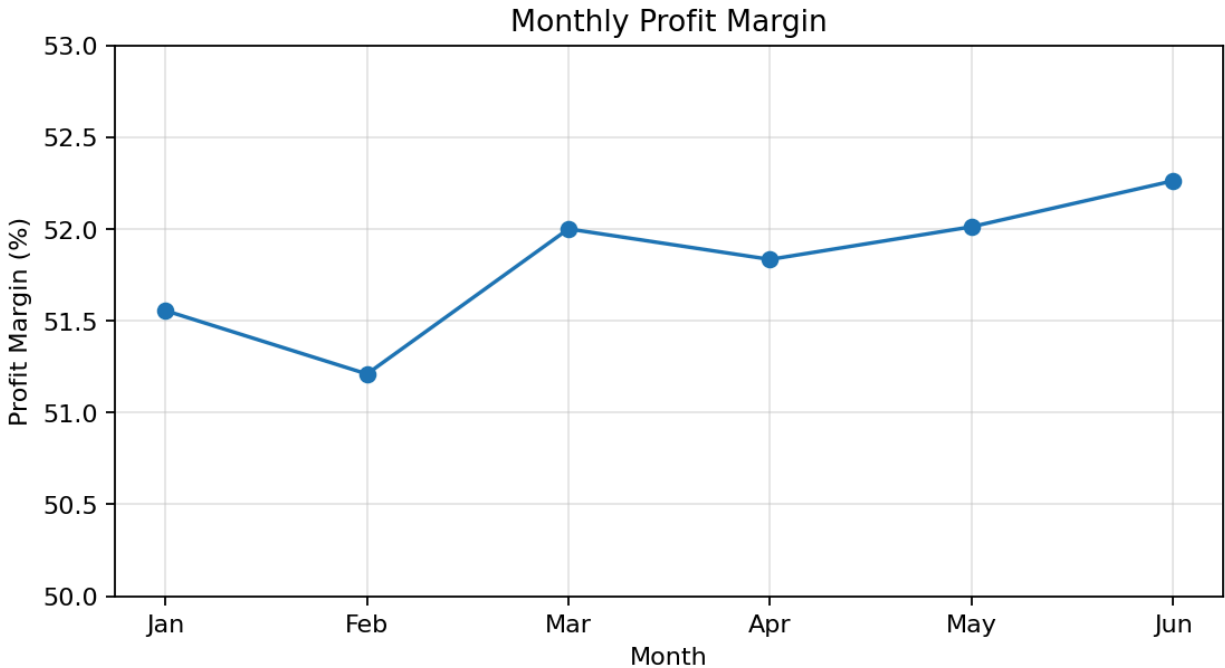


Figure 4. Monthly profit margin.

Net profit ranged from ₹33,430 to ₹37,480. February recorded the lowest profit and May the highest. The average monthly profit was ₹35,913.33.

Profit margin stayed between 51.21% and 52.26%, with an average of 51.81%. The narrow range indicates comparatively stable profitability despite changes in monthly output and costs.

Business Interpretation

- Positive profit in every month indicates that reported monthly revenue remained above reported expenses.
- May provides a useful benchmark because production, revenue and profit all reached strong levels.
- June remained profitable but showed a modest correction from the May peak.
- Stable margins suggest that revenue growth did not produce a proportionally large deterioration in reported cost efficiency.
- Monthly KPI review can help identify a sudden change in margin before it becomes a larger business issue.

The result is particularly relevant to the project problem statement because the purpose of structured data is not simply to store figures, but to make changes in financial performance visible and actionable.

3.6 Sales and Customer Findings

Sales volume, price stability and customer structure

Sales Variable	Count	Mean	Median	Std. Dev.	Range
Milk sold / transaction	1,086	4.51 L	3.00 L	2.97 L	1.50–10 L
Selling rate / litre	1,086	₹59.46	₹60.00	₹0.89	₹58–₹60
Sales amount	1,086	₹265.79	₹180.00	₹170.84	₹90–₹580

Sales Findings

The project sales analysis contains 1,086 transactions. Average milk sold per transaction was 4.51 L and average transaction value was ₹265.79. The selling rate was tightly concentrated around ₹60 per litre, with a mean of ₹59.46 and median of ₹60.

The narrow price range indicates that large changes in selling price are not the main explanation for revenue movement. Quantity sold and transaction activity are therefore more important variables for understanding sales performance.

Customer Findings

Customer Type	Number	Share
Retail	34	85.0%
Wholesale	6	15.0%
Total	40	100.0%

The customer worksheet contains 40 customer records. Retail customers form the larger group, while wholesale customers provide a smaller but distinct segment. Maintaining customer-wise quantity, frequency and payment records can help the farm understand demand and improve retention.

Business Implication

- Monitor quantity sold and transaction frequency because price variation is limited.
- Compare retail and wholesale demand separately.
- Track inactive or declining customers through regular monthly summaries.
- Use customer-wise sales information when planning milk allocation.

3.7 Feed, Veterinary and Operating Cost Findings

Cost variation and input-efficiency interpretation

Variable	Count	Mean	Median	Std. Dev.	Range
Feed quantity	1,448	13.08 kg	15.00 kg	5.25	6–20 kg
Feed total cost	1,448	₹141.75	₹120.00	₹43.89	₹96–₹224
Veterinary cost	74	₹629.05	₹600.00	₹172.21	₹350–₹850
Operating expense	724	₹982.75	₹448.00	₹1,273.89	₹55–₹4,979

Feed Findings

Average feed consumption was 13.08 kg per record and average feed cost was ₹141.75. Feed should be evaluated together with milk output. A higher feed cost is not necessarily inefficient if it produces additional milk; the useful comparison is input cost relative to output.

Veterinary Findings

The analysis contains 74 veterinary records with an average treatment cost of ₹629.05. Veterinary expenditure supports animal health, but repeated treatments should be compared with animal-wise production to identify whether health events coincide with lower productivity.

Operating Expense Findings

Operating expense entries have a relatively high standard deviation of ₹1,273.89 compared with the mean of ₹982.75. This indicates substantial variation between individual expense records. Consistent expense categories and descriptions are therefore important for reliable monitoring.

Management Implications

- Compare feed cost with milk production rather than monitoring feed cost alone.
- Maintain animal-wise health and treatment histories.
- Investigate unusually large expense entries before monthly reporting.
- Use consistent categories for labour, maintenance, utilities and other operating costs.

3.8 Consolidated Results and Business Implications

Key evidence, interpretation and management direction

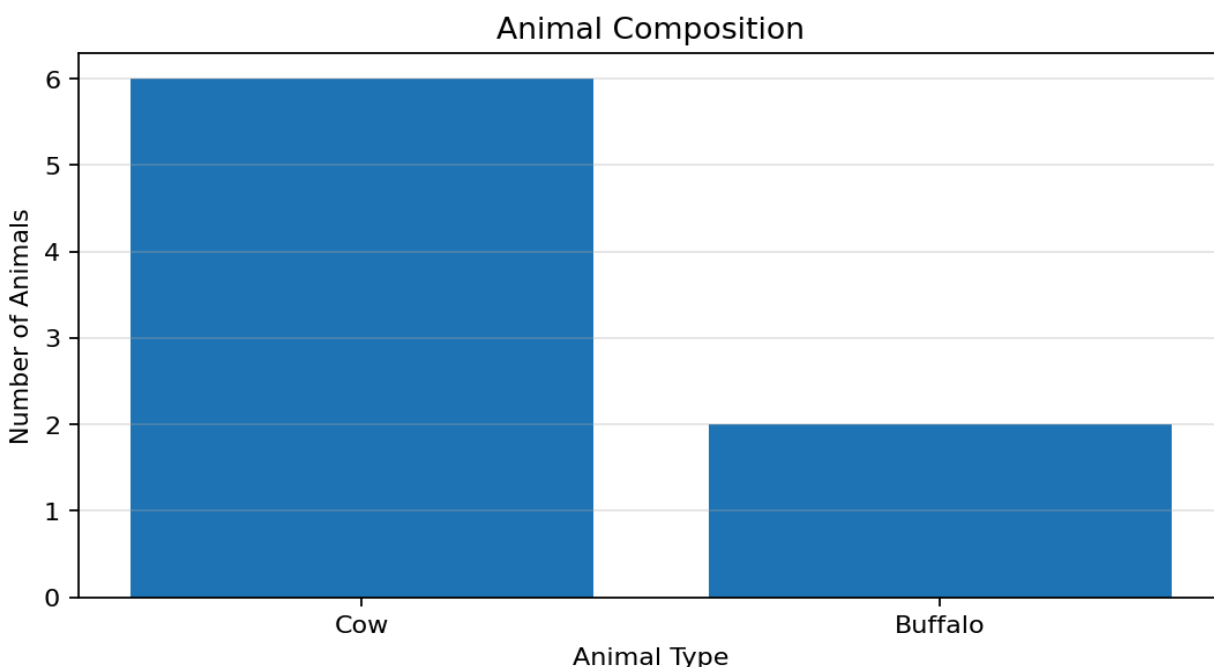


Figure 5. Animal composition in the supplied Animal_Master worksheet.

Area	Key Evidence	Interpretation
Production	1,088 L Feb → 1,201 L May	Strong recovery after February
Revenue	₹65,280 Feb → ₹72,060 May	Higher sales activity supported income
Profit	₹33,430–₹37,480	Positive in every month
Margin	51.21%–52.26%	Comparatively stable profitability
Selling price	Mean ₹59.46/L; median ₹60/L	Quantity matters more than price changes
Customers	40 records	Useful base for demand monitoring
Feed	Mean 13.08 kg; mean cost ₹141.75	Monitor input efficiency
Expenses	SD ₹1,273.89	Review large cost variations

Overall Findings

- February was the main low point for production, revenue and profit.
- May was the strongest month across the major performance measures.
- June remained strong but showed a small correction from May.
- Revenue and profit broadly followed production movement.
- Stable selling prices indicate that sales performance is driven mainly by quantity and transaction activity.
- Cost variation supports regular monitoring of feed, veterinary and operating expenses.
- The evidence supports a recurring KPI dashboard rather than one-time reporting.

Final Interpretation

The combined evidence shows that Kalka Dairy Farm maintained positive and comparatively stable performance during January–June 2025, with improvement after February. The farm's results are strongest when production, sales and costs are considered together. The analysis

provides a practical basis for monthly monitoring, animal-wise comparison and better cost control.

The findings are descriptive and cover six months, so they should not be treated as proof of long-term seasonality or causal relationships. Continued data collection over a longer period would make future trend and forecasting analysis more reliable.

Business Action Points

- Use monthly KPIs for milk, revenue, expenses, profit and margin.
- Investigate production dips promptly, especially when sales and profit also decline.
- Track feed and veterinary costs at animal level.
- Review customer quantity and transaction frequency regularly.
- Reconcile detailed records with monthly summaries before management reporting.

Originality note: This section has been newly written for the Kalka Dairy Farm project. Numerical findings are based on the supplied project workbook and the project's descriptive-statistics outputs; the proposal and mid-term report were used as project context and structure, not as text to reproduce.

4. Interpretation of Results and Recommendation –

1. Interpretation of the Results

The final analysis shows that Kalka Dairy Farm maintained a stable and profitable operating position during the six-month period from January to June 2025. The workbook reports total milk production of 6,930 litres, total revenue of ₹417,800, total expenses of ₹200,320, and net profit of ₹217,480. These figures indicate that revenue remained substantially above recorded expenses and that the farm generated a positive return throughout the study period.

The average monthly revenue was ₹69,300, average monthly expenses were ₹33,387, and average monthly net profit was ₹35,913.33. The average monthly profit margin was 51.81%, while the KPI sheet reports an overall profit margin of approximately 52.1%. The narrow monthly margin range indicates that the farm's profitability was comparatively stable even when production and expenses changed.

The results also demonstrate that the project's original business-management problem was relevant. The proposal identified manual record keeping, difficulty in accessing information, and the need for better financial monitoring. The completed workbook addresses these needs by organizing animal, production, customer, sales, feed, veterinary and expense information into separate but related worksheets, supported by monthly summaries and KPI information.

2. Overall Business Interpretation

- The farm's performance improved after the February low point, with production, revenue and profit strengthening through May.
- Milk selling rates were relatively stable, so changes in revenue were mainly associated with production and sales quantity rather than major price changes.

- Profitability remained positive in every month, showing resilience across the study period.
- Expense variation means that routine cost categorization and monitoring are important for protecting the existing margin.
- The dashboard/KPI approach converts detailed records into a management view that can support faster routine decisions.

3. Link with the Original Proposal

The final findings directly support the proposal's objectives of establishing centralized business data management and strengthening operational efficiency and decision-making. The proposed use of descriptive statistics, trend analysis, cost and profitability analysis, customer and sales analysis, and KPI/dashboard analysis was implemented in the completed study.

4. Milk Production Trend – Interpretation

Milk production was the central operational indicator in the study. Production started at 1,125 litres in January, declined to the six-month low of 1,088 litres in February, and then improved continuously through May, when it reached 1,201 litres. Production eased slightly to 1,186 litres in June. From January to June, production increased by 5.4%, showing an overall positive movement despite the temporary February decline.

5. Production-Related Business Implications

- The farm should maintain a daily production register for every animal so that changes can be identified early rather than only after a weak monthly result.
- Monthly production should be compared with animal health, feed consumption and lactation status to understand operational reasons for increases or decreases.
- May's performance can be used as a practical benchmark for future monthly monitoring, while the February result can be used as a warning level for management attention.
- The farm should avoid judging performance only from total production. Animal-wise productivity is also important because the workbook contains records for 10 animals with different production levels and statuses.

6. Animal-Wise Productivity Interpretation

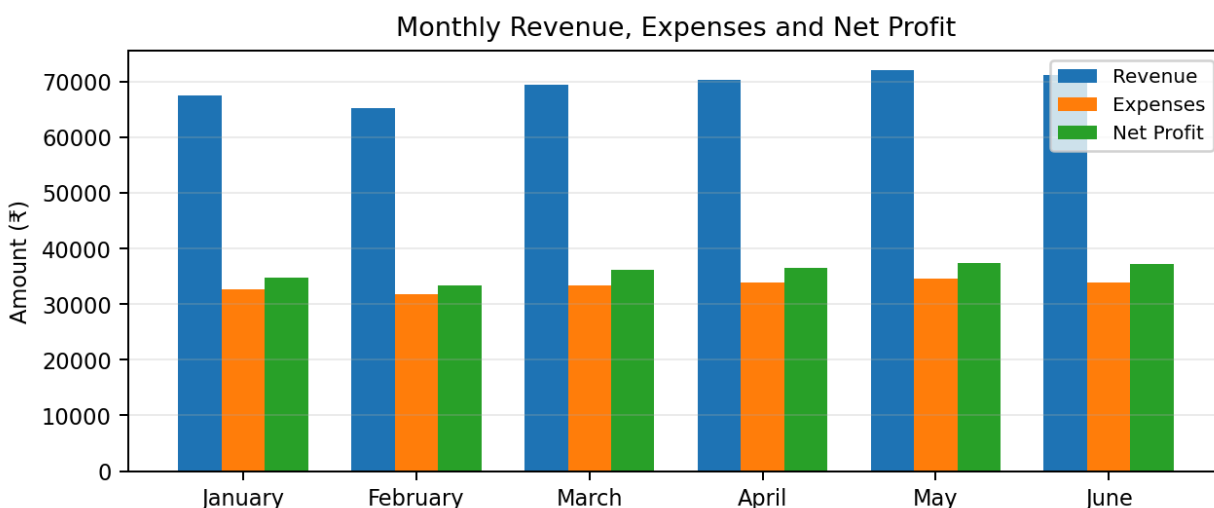
The Animal_Master sheet shows different productivity levels among lactating animals. Radha has the highest recorded average milk/day in the master data at 12.3 L, followed by Nandini at 12.0 L and Gauri at 11.5 L. At the lower end among lactating animals are Shyam at 9.5 L and Ganga at 8.8 L. Rani is recorded as pregnant and Kali as dry, so their zero production should not be interpreted as poor productivity.

This variation creates an opportunity for more targeted management. Instead of applying the same feed, health and monitoring approach to every animal, the farm can compare production with feed use and health records for each animal. This would help identify animals requiring closer observation and animals that are consistently contributing higher output.

7. Revenue, Expenses and Profitability – Interpretation

Financial performance followed the production pattern. Revenue increased from ₹67,500 in January to ₹72,060 in May, while net profit increased from ₹34,800 to ₹37,480. June showed a small correction from May: production decreased by -1.2%, revenue decreased by -1.2%, and

profit decreased by -0.8%. The movements are therefore consistent with the interpretation that stronger production periods were associated with stronger financial returns.



8. Profitability Pattern

Month	Revenue	Total Expenses	Net Profit	Profit Margin
January	₹67,500	₹32,700	₹34,800	51.56%
February	₹65,280	₹31,850	₹33,430	51.21%
March	₹69,480	₹33,350	₹36,130	52.00%
April	₹70,320	₹33,870	₹36,450	51.83%
May	₹72,060	₹34,580	₹37,480	52.01%
June	₹71,160	₹33,970	₹37,190	52.26%

Profit remained positive in all six months, ranging from ₹33,430 to ₹37,480. The monthly margin ranged from 51.21% to 52.26%, which is a relatively narrow range. This stability is important because it indicates that increases in activity did not result in a proportionately large deterioration in profitability.

9. Cost Interpretation

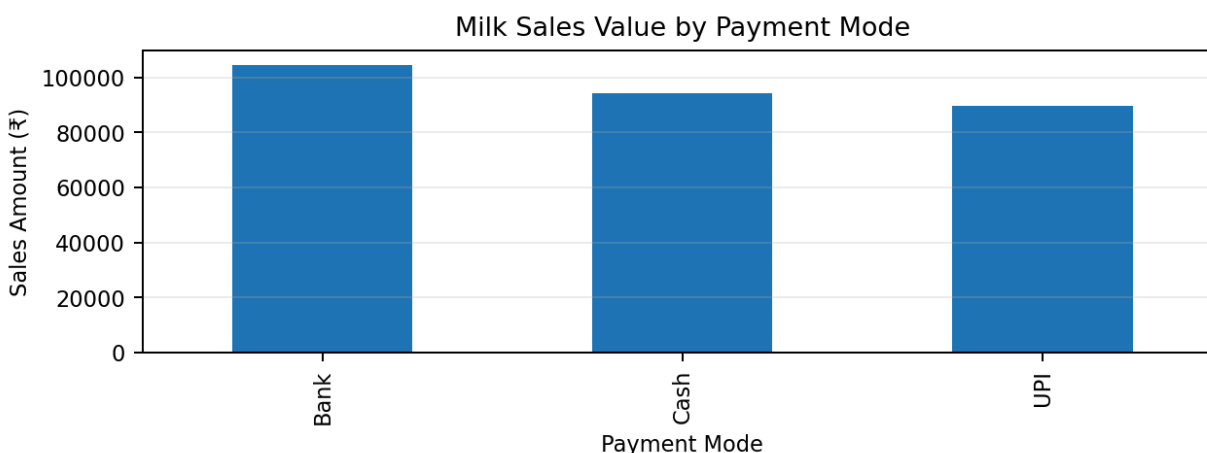
Feed is the largest recurring component in the Monthly_Summary cost structure. Average monthly feed cost was ₹18,886.67, representing approximately 56.6% of average monthly total expenses. Feed cost also moved upward during the stronger production months, which is consistent with the fact that higher production requires continuing operational inputs. The important management question is therefore not simply whether feed cost rises, but whether feed consumption and cost are producing an adequate level of milk output.

- Track feed cost together with litres of milk produced rather than monitoring feed expense alone.
- Review unusual feed-cost movements at the animal and feed-type level.
- Keep expense categories consistent so monthly comparisons remain meaningful.
- Use the monthly dashboard to compare revenue growth against total-expense growth and profit-margin movement.

10. Sales and Customer Interpretation

The Milk_Sales dataset contains 1,086 transactions. Average milk sold per transaction was 4.51 litres and the average transaction amount was ₹265.79. The selling rate was concentrated around ₹60 per litre, with a mean of ₹59.46 and a narrow observed range of ₹58–₹60. This

means that changes in sales revenue are more strongly associated with the quantity sold and transaction volume than with major price changes.



11. Customer Structure and Payment Monitoring

The workbook contains 40 customers, including 34 retail customers and 6 wholesale customers. The sales sheet also records 44 pending transactions, representing 4.05% of all transactions, with a pending amount of ₹11,854. Although most transactions are marked paid, the pending amount should be monitored because delayed collections can reduce available working cash.

Maintain a customer-wise record of quantity purchased, rate, payment mode and payment status.

Review pending payments weekly and follow up with customers before balances accumulate.

Use customer-level sales information to identify regular high-volume customers and maintain reliable service for them.

Continue monitoring the balance between retail and wholesale customers because the two groups have different quantity and pricing patterns.

12. Data Reconciliation Point

One important data-management observation from the workbook is that the Monthly_Summary/KPI level reports 6,930 litres of total production and ₹417,800 of total revenue, while the detailed Milk_Sales transaction sheet contains 4,899 litres of milk sold and ₹288,644 in transaction amounts. The supplied records do not explain the difference between these reporting levels. It may reflect differences in what is included in the monthly summary, but the reason is not explicitly documented in the workbook.

This does not invalidate the project, but it is an important future data-quality improvement. The farm should define one reconciliation rule linking production, sales, consumption/wastage if applicable, and revenue so that every litre and every rupee can be traced consistently.

13. Management Implication

The main sales opportunity is therefore not aggressive price changes. The stronger opportunity is to protect customer retention, increase reliable sales quantity, reduce pending collections, and ensure that the relationship between milk produced and milk sold is clearly documented.

14. Feed, Veterinary and Operating Expense Interpretation

The Feed_Consumption sheet contains 1,448 records, with average feed consumption of 13.08 kg per record and average total feed cost of ₹141.75. Feed usage should be evaluated together with milk output because a higher feed expense is not automatically inefficient if it supports higher productive output.

The Veterinary_Records sheet contains 74 records and the average treatment cost is ₹629.05. Veterinary spending should be linked to animal health status, treatment type and production so that recurring health issues can be identified early. Healthy animals are important not only for welfare but also for maintaining stable production.

The Expenses sheet contains 724 operational expense records across categories such as feed purchase, labour, veterinary, electricity, maintenance, transport and miscellaneous expenses. The descriptive analysis reports a mean expense amount of ₹982.75 per record and a high standard deviation, indicating substantial variation among individual expense entries.

15. Recommendations for Cost and Operational Control

- Introduce a daily digital record routine covering milk production, feed, sales, payments, veterinary activity and expenses.
- Review feed cost and milk output together at animal level and monthly level; investigate unusual changes rather than reducing feed blindly.
- Maintain a preventive animal-health calendar and record follow-up visits so recurring treatment patterns can be identified.
- Categorize every operating expense consistently and review monthly totals against the previous month.
- Separate routine operating expenses from unusual or one-time expenses so management can distinguish normal cost movement from exceptional events.

16. Business Data Management Recommendations

The central recommendation from the project is to continue moving from manual, register-based management toward a structured digital business data system. The proposal originally identified the need to centralize animal, milk, customer, sales, feed, healthcare and financial information. The completed workbook demonstrates that this structure is practical for the farm.

- Use one standardized workbook/database structure with fixed field names and validation rules.
- Assign a unique ID to every animal, customer, sale, feed record, veterinary record and expense.
- Protect formula cells and maintain a separate raw-data area from summary/dashboard areas.
- Back up the file regularly and keep monthly versions so historical records are not lost.
- Use the KPI dashboard as the first page for monthly management review, followed by detailed sheets only when investigation is needed.

17. Recommended KPI Dashboard

KPI	Current Value	Management Purpose
Total Milk Produced	6,930 L	Track production scale
Total Revenue	₹417,800	Track sales income
Total Expenses	₹200,320	Control operating cost
Net Profit	₹217,480	Monitor financial return
Profit Margin	52.1%	Protect profitability
Customers	40	Monitor customer base
Animals	10	Monitor livestock base

18. Final Recommendations and Action Plan

The final recommendations are designed to be practical for a small dairy farm. They focus on improving the existing system rather than introducing unnecessary complexity. The analysis shows that the farm is already profitable, so the priority is to protect this performance while improving visibility, consistency and control.

No.	Recommendation	Action	Frequency
1	Daily digital record keeping	Record milk, feed, sales, payments, veterinary activity and expenses every day.	Immediate
2	Monthly KPI review	Review production, revenue, expenses, profit, margin, customers and animal performance.	Monthly
3	Production monitoring	Compare each animal's production with feed use and health status.	Weekly/Monthly
4	Expense control	Categorize and review unusual expenses; compare feed cost with milk output.	Weekly/Monthly
5	Payment follow-up	Track the 44 pending sales transactions and ₹11,854 pending amount.	Weekly
6	Data reconciliation	Document how production, sales quantity and revenue are reconciled.	One-time + Monthly
7	Backup and validation	Use standard formats, unique IDs, validation rules and regular backups.	Continuous

19. Presentation and Legibility of the Report

The final report should present the analytical story in the same sequence used by the supplied reference final report: first explain the interpretation of the results, then connect each interpretation to a practical recommendation, and finally present the strategic outlook. For Kalka Dairy Farm, the report should remain focused on the farm's two original problem statements—organized business data management and improved operational/financial decision-making.

Use clear numbered headings and sub-headings so the evaluator can locate Interpretation, Findings, Recommendations and Conclusion quickly.

Keep tables compact and place the unit in every numerical heading or column, such as L, ₹ and %.

Use charts for trends and comparisons, with readable axis labels, legends and titles.

Avoid overcrowding pages; maintain consistent margins, spacing and alignment.

Use one professional font family throughout the report and maintain a clear hierarchy between title, heading and body text.

Bold the most important findings and recommendations, but avoid excessive bolding.

Ensure every chart and table has a descriptive title and is directly discussed in the surrounding text.

Keep the final report consistent with the proposal and mid-term report terminology so that the progression of the project is easy to follow.

20. Conclusion

The six-month analysis provides evidence of a stable and improving business performance at Kalka Dairy Farm. Production increased overall from 1,125 litres in January to 1,186 litres in June, after reaching a low of 1,088 litres in February. May was the strongest month for production, revenue and net profit. Across the period, the farm generated ₹217,480 net profit on ₹417,800 reported revenue, with an overall margin of approximately 52.1%.

The most important managerial lesson is that the farm does not require a complicated analytical system to gain value from data. A disciplined digital record structure, regular KPI review,

animal-wise monitoring, categorized expense tracking and customer-payment follow-up can convert the existing records into a reliable management system.

Overall, the project successfully connects the original proposal, the mid-term analytical work and the final business interpretation. The recommended approach is to continue using structured Excel/Python-supported analysis and a simple dashboard, while improving data reconciliation and routine record discipline. This will help Kalka Dairy Farm preserve its current profitability, identify operational issues earlier and make better evidence-based decisions for future growth.

I am also providing Excel sheet and colab notebook for reference

[Kalka dairy farm data](#)

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